

Hillary Sanders

Highlights:

- 12 years of experience in machine learning research and engineering
 - 8 years experience in leading ML (neural network) research & eng teams
- Lead researcher / engineer for multiple billion-sample+ deep learning models
- Author of deep learning book chapters & other publications

EXPERIENCE

AE Studio

Alignment Scientist
2026-Present

- Developing methods to better align frontier AI models.

Heroku

Principal Member of
Technical Staff, AI
Engineer & Architect
2024-2026

- Leveraged tools like Triton, TensorRT-LLM, VLLM, and PyTorch to deploy cost-performant, resilient LLMs & diffusion models to production at scale.
- Developed custom agentic flows to leverage the power of LLMs in complex tasks (RAG, tools, graph-based action flows).
- Developed foundation model API for Heroku users

Opendoor

Senior Research
Scientist
2021-2023

- Designed & conducted experiments on Opendoor's core model (house-price prediction neural network) resulting in over \$50M of total expected profit gains
- Led team of 6 researchers and engineers to re-engineer ML prod training and experimentation pipeline to improve usability, reproducibility, and uptime

Sophos

Senior ML Researcher
2019 - 2021
Senior Software
Engineer
2018- 2019
Data Scientist
2017- 2018

- Developed catastrophic forgetting mitigation techniques to avoid retraining models from scratch, allowing for iterative fine-tuning over time with appx. 98% reduced forgetting effects, and ~10x reduced model training costs
- Developed novel model compression technique (variance-based Taylor series approach), resulting in appx. 30% less reduction in accuracy after 90% weight removal compared to the next-best method
- Owned research & maintenance for two deep learning multi-million parameter classification models, trained on billions of document samples
- Led engineering team for our Endpoint Detection Response implementation (Docker, Elasticsearch, 20 separate models); also designed and maintained the production databases to support these (Elasticsearch)
- Co-designed the team's main database (Redshift) with billion+ row tables; implemented a pipeline to create aggregation tables for ML training
- Crafted algorithms for developing better labels in our data

RecipeStasher

Founder / Full-Stack
2016 (Sep-Dec)

- Designed and built a recipe website. Key innovation over state-of-the-art was natural language recipe parser which could automatically convert unit and serving conversions across both ingredients and instructions

Premise Data Corporation

Data Scientist
2013 - 2016

- Developed data processing pipeline for our main product (inflation indices), as well as various other analytic pipelines (worker reliability, etc)
- Developed and implemented an optimal survey volume formulation to minimize expected variance in our inflation indices

PUBLICATIONS

2021 **arXiv** - Magnificent Minified Models: approaches to model compression

2021 **arXiv** - Catastrophic Forgetting in the Context of Model Updates

2018 **No Starch Press (book)** - Authored the chapters of "Malware Data Science" that focus on deep learning theory and methods

2018 **IEEE** - A Deep Learning Approach to Fast, Format-Agnostic Detection of Malicious Web Content

2017 **BlackHat** - Garbage in, Garbage Out: How purportedly great ML models can be screwed up by bad data



EDUCATION

UC Berkeley

BS Statistics

BS Env. Econ & Policy

2009-2013

- Graduated Cum Laude
- Selected into competitive undergraduate research program twice:
 - Researched visualizing & tracking changes in legislative documents via NLP with graduate student Mark Huberty
 - Researched how liability affects international technology transfer

SOFTWARE

- Daily user of **Python, ec2, Pytorch, PostgreSQL, Bash, AWS, Git**
- Frequent user of **Docker, Spark, Snowflake, Airflow**